

RAG 메뉴얼



Milvus 버전

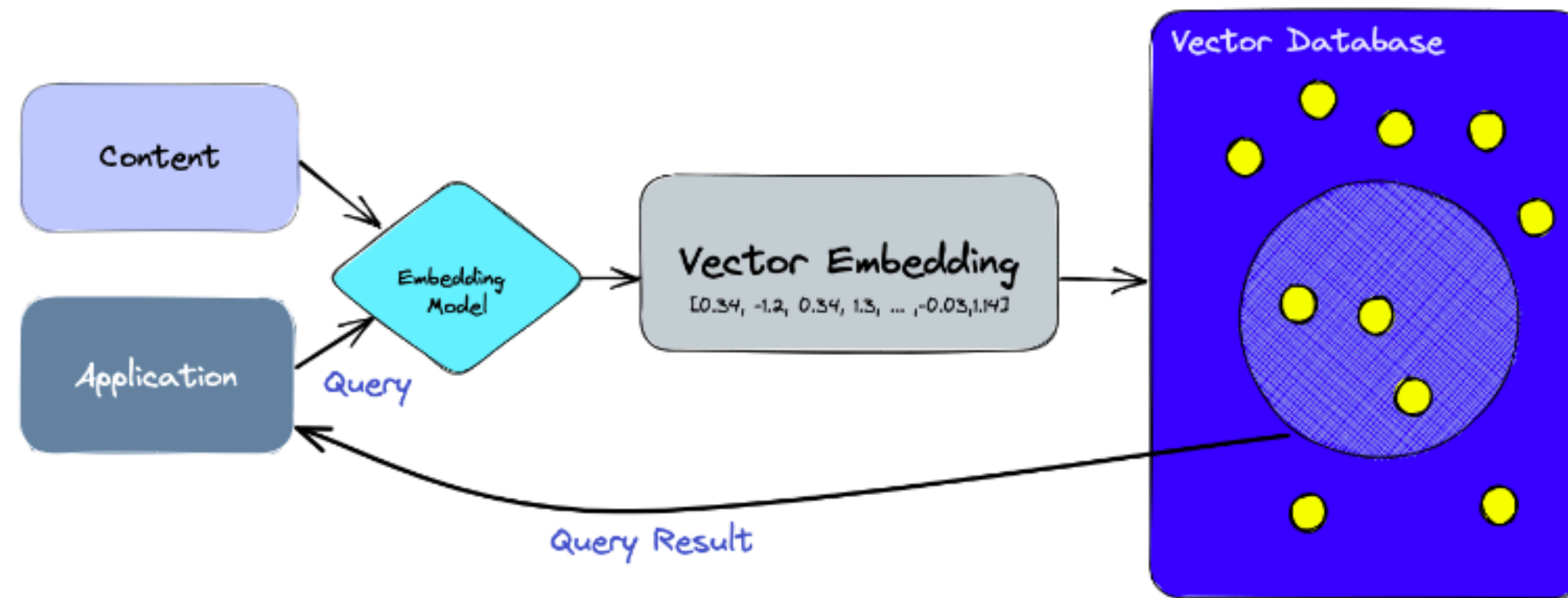
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VectorDB 개념



1. **데이터 관리:** 삽입 삭제 업데이트와 벡터 인덱스를 사용하는 것보다 벡터 데이터를 쉽게 관리하고 유지할 수 있다.
2. **메타데이터 저장 및 필터링:** 벡터 항목 관련 메타데이터를 저장할 수 있다. 필터를 이용하여 DB쿼리 가능
3. **확장성:** 데이터 양이 많아질 수록 분산 및 병렬 처리됨 (샤딩, 수평 분할)
4. **실시간 업데이트:** 실시간 데이터 업데이트를 지원(동적으로 변경 가능)
5. **백업 및 수집:** 모든 데이터를 백업하는 일상적인 작업 처리(배치), Collections 형태로 백업 가능(NoSQL 특성)
6. **에코시스템 통합:** Spark, 태블로, 그라파나 같은 분석, 시각화 툴들과 생태계 연동 가능

VectorDB 비교



Chroma



Vector Database Feature Comparison Matrix

Vector DB Comparison
Vector DB Comparison is a free and open source tool from VectorHub to compare vector databases.

 superlinked.com

Source: <https://superlinked.com/vector-db-comparison>



Azure Cosmos DB



Postgres + PG Vector



임베딩 모델

서버 경로: /mnt/nvme01/huggingface/models/ + (모델명)

<u>언어 지원</u>	<u>모델명</u>	<u>차원(Dimension)</u>	<u>모델 사이즈</u>	<u>Max Seq Length (토큰)</u>
영어만	Alibaba-NLP/gte-base-en-v1.5	768	137M	8192
	Alibaba-NLP/gte-large-en-v1.5	1024	434M	8192
	BAAI/bge-base-en-v1.5	768	109M	512
	BAAI/bge-large-en-v1.5	1024	335M	512
다국어 (영어 / 한국어 지원)	Alibaba-NLP/gte-multilingual-base	768	305M	8192
	BAAI/bge-m3	1024	-	8192
	BAAI/bge-multilingual-gemma2	4096	9.24B	8192

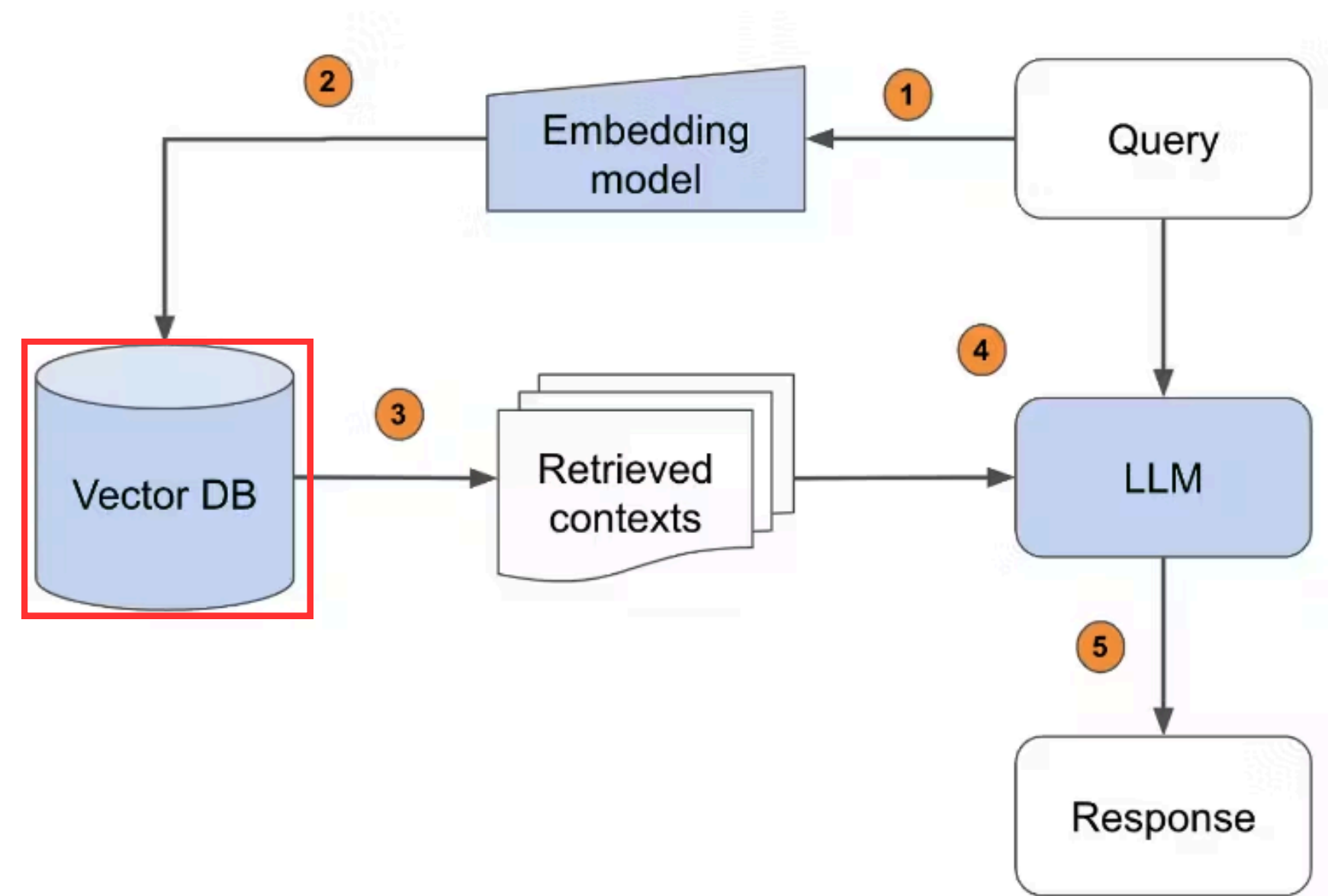
Milvus 소개



Source: <https://milvus.io/docs/ko/quickstart.md>

RAG 구축 - 인덱싱 방법

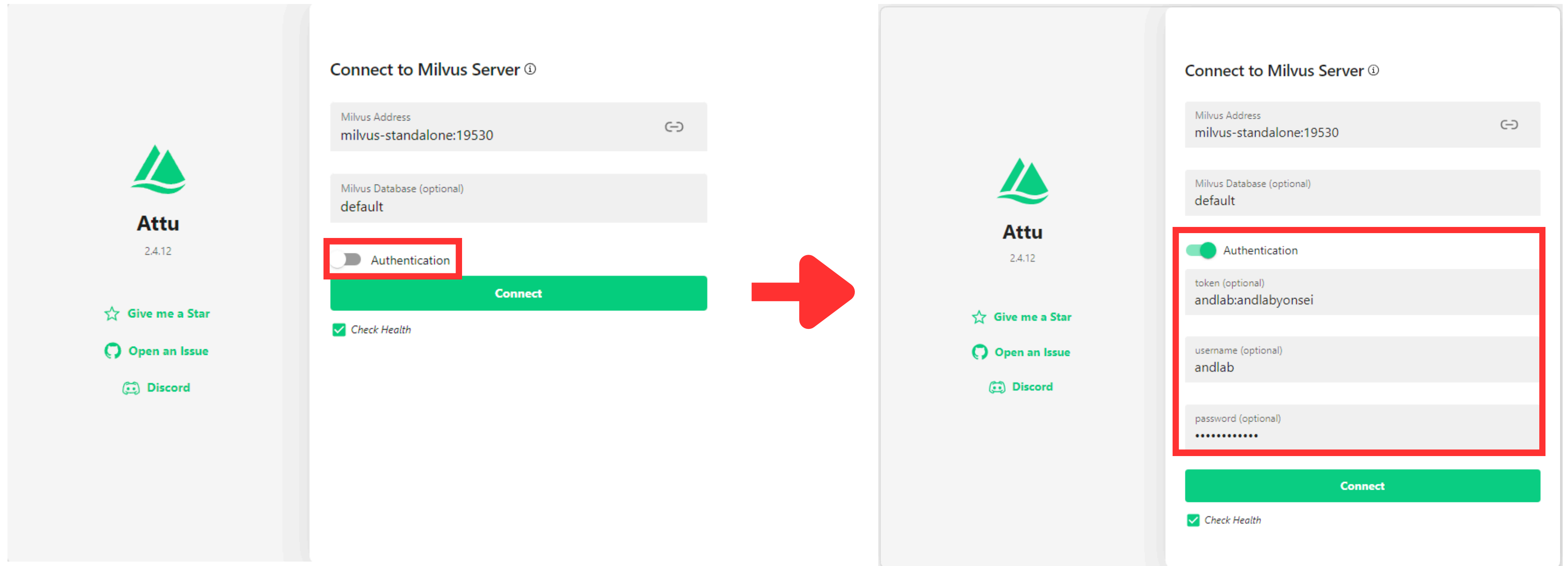
분류	구축 기반	인덱싱 종류	유사도 측정 항목
In-Memory	CPU + RAM 메모리	FLAT (IVF / IVF-SQ8 / PQ)	Euclidean Distance (L2) / Inner Product (IP)
		HNSW	
		SCANN	
On-Disk (현재 서버에 사용 중)	SSD / NVMe 메모리	DiskANN	Euclidean Distance (L2) / Inner Product (IP) / Cosine
GPU	그래픽 카드 메모리	GPU_CAGRA	Euclidean Distance (L2) / Inner Product (IP)
		GPU_IVF_PQ / GPU_IVF_Flat	
		GPU_BRUTE_FORCE	



Source: <https://milvus.io/docs/v2.4.x/index.md?tab=floating>

Milvus Attu 접속

접속 사이트 주소(URL): <http://165.132.192.52:8000>



Login

token(optional): andlab:andlabyonsei

username(optional): andlab

password(optional): andlabyonsei

Milvus Attu 접속

접속 사이트: <http://165.132.192.52:8000>

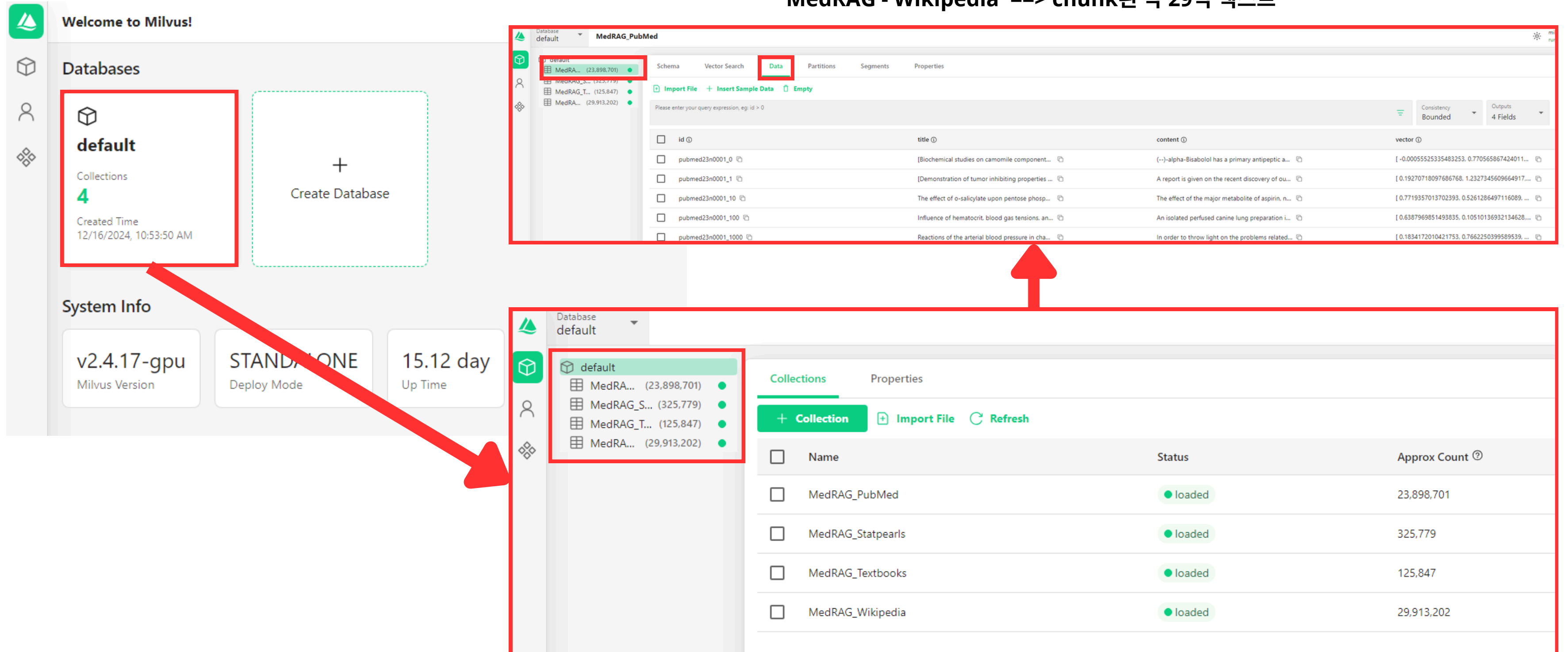
구축된 RAG 문서 리스트

MedRAG - PubMed ==> chunk된 약 23억 텍스트

MedRAG - StatPearls ==> chunk된 약 32만 텍스트

MedRAG - Textbooks ==> chunk된 약 12만 텍스트

MedRAG - Wikipedia ==> chunk된 약 29억 텍스트



The image shows the Milvus Attu web interface. On the left, the 'Databases' section shows a 'default' database with 4 collections, created on 12/16/2024 at 10:53:50 AM. Below this, 'System Info' shows version v2.4.17-gpu, deployment mode STANDALONE, and uptime of 15.12 days. A red arrow points from the 'default' database entry to the 'Collections' tab of the 'MedRAG_PubMed' database. The 'MedRAG_PubMed' database details show a list of collections: MedRAG_PubMed (23,898,701), MedRAG_Statpearls (325,779), MedRAG_Textbooks (125,847), and MedRAG_Wikipedia (29,913,202). A red arrow points from the 'MedRAG_PubMed' collection entry to the 'Data' tab, which displays a table of document chunks with columns: id, title, content, and vector.

id	title	content	vector
pubmed23n0001_0	[Biochemical studies on camomile component...	(-)-alpha-Bisabolol has a primary antieptic a...	[-0.0005525335483253, 0.770565867424011...
pubmed23n0001_1	[Demonstration of tumor inhibiting properties ...	A report is given on the recent discovery of ou...	[0.19270718097686768, 1.2327345609664917...
pubmed23n0001_10	The effect of o-salicylate upon pentose phosp...	The effect of the major metabolite of aspirin, n...	[0.7719357013702393, 0.5261286497116089, ...
pubmed23n0001_100	Influence of hematocrit, blood gas tensions, an...	An isolated perfused canine lung preparation i...	[0.6387969851493835, 0.10510136932134628...
pubmed23n0001_1000	Reactions of the arterial blood pressure in cha...	In order to throw light on the problems related...	[0.1834172010421753, 0.7662250399589539, ...

PyMilvus 접속

```

RAG_search.py
Code > preprocessing > RAG_search.py > ...
Run Cell | Run Below | Debug Cell
1  # %%
2  from pymilvus import connections, Collection
3  from sentence_transformers import SentenceTransformer
4
5  model_folder = '/mnt/nvme01/huggingface/models/'
6  embed_name = model_folder + "Alibaba-NLP/gte-large-en-v1.5"
7  encoder = SentenceTransformer(embed_name, trust_remote_code = True)
8  connections.connect(host = "localhost", port = "19530", token = "andlab:andlabyonsei")
9  collection_lists = ['MedRAG_PubMed', 'MedRAG_Wikipedia', 'MedRAG_Statpearls', 'MedRAG_Textbooks']
Run Cell | Run Above | Debug Cell
10 # %%
11 rag_contents = []
12 for names in collection_lists:
13     collection = Collection(name = names)
14
15     res = collection.search(
16         data = encoder.encode(["tell me about chicken-pox related information"]),
17         anns_field = "vector",
18         param = {"metric_type": "IP",
19                 "search_list": 8},
20         limit = 1,
21         output_fields = ["id", "content"],
22     )
23     rag_contents.append(res[0][0].content)
24
25 print(rag_contents)

```

임베딩 코드

쿼리 및 검색 코드

사용자 계정별>

먼저 “pip install pymilvus==2.4.12 sentence-transformers” 설치해야함

RAG 검색 방법:

1. 임베딩 모델 설정
2. 인코딩할 쿼리 입력

```

rag_contents
[3]  ✓  0.0s
... ['Varicella (chicken pox) is a highly contagious disease which is caused by Varicella Zoster Virus (
    'Chickenpox occurs in all parts of the world. In 2013 there were 140 million cases of chickenpox an
    'Chickenpox or varicella is a contagious disease caused by the varicella-zoster virus (VZV). The vi
    'EPIDEMIOLOGY AND CLINICAL MANIFESTATIONS Chickenpox Humans are the only known reservoir for VZV. C

```

*Thank
you!*